

TECHNOLOGIA 2.0

Round 2 · Operation Phoenix

MISSION 3 OF 4

System Expansion

Suggested time: ~15 minutes | Deliverable: Architecture Version 2

NimbusCart's pilot launch went well. With a major regional festive sale coming up, the client wants the platform ready for a large, short-term surge in traffic — and a better shopping experience to go with it.

New / Changed Client Requirements

- Buyers should see personalized product recommendations based on their browsing and purchase history.
- The platform must stay responsive under a sudden spike in read traffic during the sale window.
- Product images and static assets should load quickly for buyers across different regions.
- The operations team wants basic monitoring and alerting in place before the sale, to catch issues early.

Constraints Update

- The 9,500 budget unit monthly cap still applies to the **combined** architecture — Mission 1 selections plus anything you add now.
- You may revisit and change an earlier (Mission 1 or 2) choice if a new requirement genuinely calls for it — but be ready to justify the change.
- Not every new requirement needs a brand-new component; some may be satisfied by adjusting how an existing one is used.

Task: Extend to Architecture Version 2

- Identify which new requirements need a new component from the Technology Inventory, and which can be met by adjusting your existing architecture.
- Add the components you choose to your Mission 2 diagram to produce Architecture Version 2.
- Keep the original request flow intact — you're extending the system, not starting over.
- Note briefly, next to each addition, which new requirement it addresses.

Deliverable

A completed **Architecture Version 2** diagram, extending your Mission 2 design with the components needed to satisfy the new requirements, and a brief note on why each was added.

Evaluation Focus

Problem Solving and Decision Making • Software Architecture Design